

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: on a simulated cortical sheet it constrains at most 10 sites, chaining through overlapping patches extends it to 0.312 mm, and across distant regions the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies

these implications; one model satisfies all connecting hypotheses at once, showing consistency, not cortical realization. The conditions yield a disconfirmable comparison: at matched phase coherence and single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. Choosing the overlapping regions by what each decodes makes the unity correlate operational relative to a pre-registered decoder and criterion. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent

15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which; each result
23 names a measurement that phase data alone cannot supply.

24 The objection that synchrony does not fix content is older than this work.
25 Critiques of the temporal binding hypothesis argued that synchrony, where
26 it occurs, cannot say which features belong together or what they jointly
27 represent: that grouping has to be computed before synchrony could sig-
28 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
29 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
30 and a pigeonhole count. What formalization adds to the informal argument
31 is quantitative and bookkeeping content. It gives the rate at which a co-
32 herence bound on content weakens with population size, the population at
33 which it stops constraining anything and how far chaining through overlap-
34 ping patches extends it, the diameter within which one contracting map can
35 reconstruct, and the number of codes a separated family of states requires.
36 It also gives a complete inventory of the premises a synchrony-to-content
37 argument consumes, each of which names a measurement, together with one

38 model in which all of them hold at once.

39 The familiar noisy mean-field oscillator supplies a stationary coherence
40 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991); we add a minimum
41 cost for changing coherence quickly. A Lean 4 development checks the impli-
42 cations, and a joint witness satisfies the eight connecting hypotheses of the
43 formal composition in a specified model. It proves neither that cortex satisfies
44 those hypotheses nor that the glued state is experienced; that identification
45 needs its own endpoint.

46 The empirical question is which measurements could support or discon-
47 firm the proposed connections. A phase resultant can miss a locally ordered
48 winding field; stationary phase distributions can leave the coupling mecha-
49 nism unidentified; and coherent adaptation can lose alignment with a desig-
50 nated content template. We therefore distinguish stationary branches from
51 finite observed onset, and compare prespecified field and connectivity onset
52 models on held-out emergence recordings. The comparison tests a specified
53 spatial prediction, not the presence of experience. Section 10.2 states the
54 comparison that bears on conscious content directly.

55 Sections 2–5 develop the three requirements; Section 6 applies their mea-
56 surement obligations to a digital candidate. Section 7 states what the formal
57 composition establishes, and Sections 8–9 examine observables and the cor-
58 tical test. Table 1 lists the connecting hypotheses the article argues from;
59 Supplemental Material gives the full chain, derivations and controls. For a
60 theoretical article the Methods are the definitions, proofs and simulations.

Each result is stated in the section that uses it, Supplemental Material Table S1 maps it to its Lean declaration, and every simulation’s model, parameters and controls are given in Supplemental Material. The Results are the theorems of Sections 3–5 and the numerical controls of Section 8.

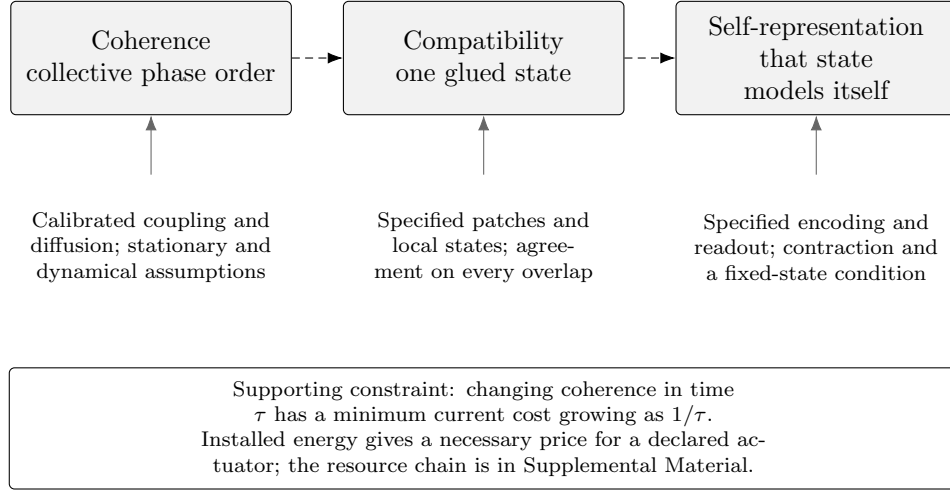


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

64

2 Limits of phase order

Several limits hold without the cortical hypothesis. Equal phases can coexist with incompatible local descriptions; phase order constrains shared content only through encoders related on overlaps. A winding field can have zero

69 global resultant while remaining locally ordered. A constant readout can have
70 a unique fixed point without reconstructing its input. A region encodes only
71 the states it distinguishes, and by a declared deadline a site can report only
72 changes inside its causal past. These limits reflect familiar phase oscillator
73 and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008;
74 Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one
75 beside the condition or measurement it motivates.

76 **3 Coherence: physical coupling and its cost**

77 **3.1 Dynamics and the stationary threshold**

78 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

79 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
80 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
81 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
82 quency heterogeneity and diffusion are distinct parameters. The stationary
83 calculation below uses identical frequencies and this scalar coupling. Ap-
84 plying it to a spatial kernel requires an aggregation rule and a check of the
85 resulting approximation.

86 For identical frequencies in their rotating frame, the positive normalized

87 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

88 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 89 mal derivation uses the absence of stationary current for a periodic gradient
 90 drift to obtain this density, then verifies normalization and self-consistency.
 91 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 92 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 93 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

94 Figure 2 shows this stationary branch.

95 Linearization about the uniform density gives rate $K/2 - D$ for the first
 96 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 97 exactly above threshold. About the coherent density, the quadratic form
 98 is strictly dissipative transverse to collective rotation, whose rate is zero.
 99 These linear estimates concern the limiting equation. Relating them to finite
 100 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 101 tion; exchange symmetry alone does not establish the needed independence
 102 (Sznitman, 1991).

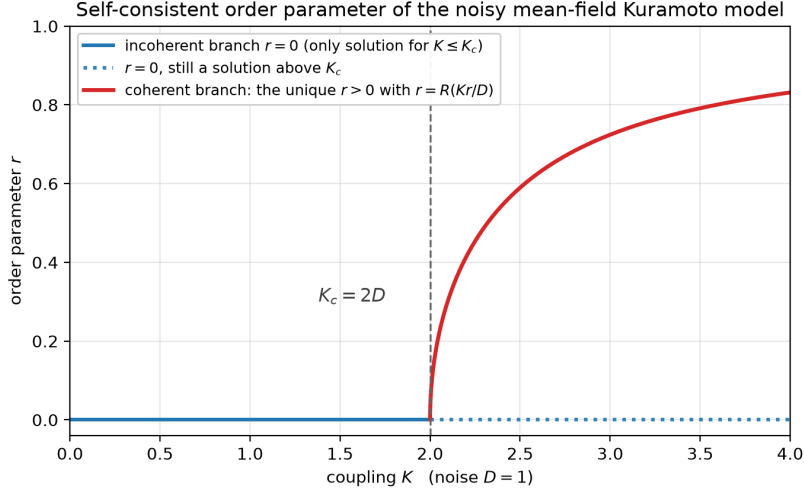


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

103 3.2 A speed limit on changes of coherence

104 One thermodynamic statement bears on coherence directly. For a smooth
 105 positive periodic phase density with probability current J and diffusion D ,
 106 the current functional $\sigma_J = \int J^2 / (D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

107 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
 108 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
 109 has no positive floor. Read the other way, a current budget C sets a minimum

110 duration $\tau \geq [\Delta \arcsin r]^2 / (DC)$ for a given change. This bears on ignition-
 111 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
 112 2014), and on recovery latency after anaesthesia: each has a floor set by the
 113 current available to drive it. The bound is a heat or power statement only
 114 when J is identified with a physical flux, a reservoir at a stated temperature
 115 is specified and local detailed balance holds for the steps concerned; only
 116 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
 117 Shenker, 2000). Installed energy supplies a separate, necessary price: for
 118 a declared actuator whose response modes carry installation prices, mean
 119 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
 120 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$, and no cortical value of κ
 121 is known. Supplemental Material gives both derivations, the register and
 122 feedback budgets, and the sensitivity of the metabolic comparison to its
 123 assumptions.

124 **4 Compatibility: from coordinated activity** 125 **to a global description**

126 **4.1 Why common timing is insufficient**

127 The formal setting is a sheaf: each patch of the substrate carries local de-
 128 scriptions, and restriction maps cut a description on a patch down to any
 129 smaller patch. A *section* is a description on one patch, and a *global section* is

130 a description on the whole substrate. For overlapping patches U and V , com-
131 patibility means that restricting either local description to $U \cap V$ gives the
132 same object. Descriptions can agree on shared quantities while contributing
133 different information elsewhere. Compatible sections of the specified sheaf
134 have a unique global extension. On a finite cover, local finite spatial mea-
135 sures agreeing as measures on every overlap also have a unique finite-measure
136 extension whose sheaf image is the glued section. No mass-one normalization
137 is imposed.

138 In the cup example, visual and reaching descriptions must express a
139 shared quantity, such as cup location, in common coordinates. Restriction
140 retains that quantity while dropping details particular to each description.
141 Agreement permits surface appearance and hand configuration to contribute
142 to one situation. The mathematical counterexample assigns the same phase
143 to every patch while two local mass profiles disagree on a shared site. Those
144 profiles have no common extension. Because they are chosen independently
145 of phase, synchrony places no constraint on their disagreement.

146 This distinction limits the biological interpretation. Receptive territories
147 and content domains must be related to spatial patches, and the descriptions
148 must support the required extension. Matching overlap marginals of de-
149 coded probability laws need not determine a unique global joint distribution
150 (Abramsky & Brandenburger, 2011).

151 The cover is itself a choice, and the construction constrains it in one way
152 only: synchronized patches agree where they overlap. Where they do not

153 overlap there is nothing to compare, so a cover by pairwise disjoint patches
154 satisfies that obligation for *every* assignment of local descriptions and phases
155 and the gluing theorem still returns a unique global section, which is the
156 tuple it was handed. Membership of the class then reports nothing about the
157 system beyond the patches covering it. What a gluing claim asks of a cover
158 is therefore a non-degeneracy the theorem does not assume: that distinct
159 patches meet, and that what they share is a quantity fixed independently of
160 the agreement it is being used to establish.

161 The cup example exposes a second gap, between two kinds of overlap. The
162 sheaf is built over a spatial substrate, so its patches overlap where they share
163 sites. Visual and motor territories that agree on cup location overlap in what
164 they describe, not in where they lie, and the construction supplies no relation
165 between the two. One route recasts the construction over content variables:
166 each region becomes a partial decoder whose domain is the set of variables
167 it estimates, and two regions overlap exactly where they estimate a common
168 variable. The gluing argument needs only a cover and restriction maps, so
169 it applies formally to such a cover, but no result here is stated for one, and
170 which variables and decoders form it is an empirical question. One principle
171 selects such a cover without presupposing the agreement it is used to test.
172 Assign each region the variables that can be decoded from its activity above
173 a declared criterion, on data held out from the agreement test, and let two
174 regions overlap on exactly the variables decodable from both. Decodability
175 is fixed by what each region carries before any comparison is made, so the

cover cannot be chosen to make agreement come out. It is non-degenerate exactly when its nerve is connected, meaning that every region is linked to every other through a chain of shared decodable variables, and recordings can fail that condition. The principle has three inputs, the candidate variables, the criterion and the decoder class, and each has a rule that does not consult agreement.

The candidate variables are those the experiment manipulates and the task makes behaviourally relevant, fixed before recording. Data-driven alternatives, such as directions of maximal shared variance between two regions, select for the agreement under test and are excluded. The criterion becomes a statistical one: a region encodes a variable when its cross-validated decoding exceeds a permutation null at a pre-registered level, and results are reported across a declared range of levels. The level need not be exact, because two accurate decoders agree automatically (Eq. (13)); the statistic of Section 10.2 is agreement in excess of what each region’s accuracy implies, which remains valid at any admitted level. The decoder class is bounded on both sides. Unrestricted nonlinear decoders recover a variable from any region carrying information about it, so nearly every pair of regions overlaps and the cover again constrains nothing. The positive bound of Section 4.2 requires encoders with a declared Lipschitz constant L , which enters δ_r directly, so the class should consist of L -Lipschitz maps; linear readout, the usual operationalization of information explicitly available to downstream neurons (DiCarlo & Cox, 2007; Kriegeskorte & Diedrichsen, 2019), is one such class. Within the

199 admitted family, the decoder and the distance in which content discrepancy is
 200 measured are chosen by held-out decoding accuracy and never by agreement.
 201 Neither is neutral: which similarity measure best describes neural activity
 202 differs across tasks (Bobadilla-Suarez et al., 2020). What remains is that the
 203 cover is the experimenter’s. Whether cortex realizes unity over the chosen
 204 variables is the empirical claim under test, and no choice of decoder settles
 205 it. The correlate is therefore operational relative to these pre-registered in-
 206 puts, and carrying the formal results to such a cover is the recasting sketched
 207 above.

208 **4.2 When coherence does constrain content**

209 A declared relation between phase and content gives a positive result. Let
 210 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 211 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 212 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance,
 213 meaning that an encoder’s output moves by at most L times the chord
 214 between two input phases, and require $e_{i,x} = e_{j,x}$ wherever patches i and
 215 j overlap. Encoders may vary across sites; their equality on overlaps is a
 216 supplied relation between descriptions of the shared quantity. Write $z_k = e^{i\theta_k}$
 217 and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The phasors’ squared deviations from their
 218 mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$, and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$
 219 gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair. Consequently every overlap

220 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (5)$$

221 Here discrepancy is the norm of the difference, and r and N are read on
222 whichever population the bound is applied to. That choice is what decides
223 whether the estimate says anything. A patch carries an order parameter
224 of its own, and reading the bound there carries that patch's resultant and
225 that patch's site count, so a large population observed through small patches
226 is constrained by the small count. Read instead on the whole array, the
227 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
228 from an average over N sites. The order cannot be improved in general:
229 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
230 observations agree exactly at either scale. A shared scalar encoder is a special
231 case. Dropping the overlap relation permits disagreement even at perfect
232 locking. A common nonexpansive linear projection preserves agreement, but
233 agreement of a projection does not establish agreement of the full vector.

234 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
235 constrains anything only where the right-hand side falls below that. At the
236 sheet's own patch order, the bound is informative for at most 10 sites. Three
237 conditional results decline the step that puts $\sqrt{N}$ there. A pair count bounds
238 the fraction of disagreeing pairs at any population size but names no pair.
239 A chain across the sites that overlapping patches share compares distant

territories through the patches between them, with error growing linearly in hops rather than with population: on the same sheet it stays informative for 19 hops, 0.312 mm. A spectral gap removes the population count at a frequency-locked configuration, given a declared algebraic connectivity of the coupling graph. Territories whose agreement unity concerns lie centimetres apart, beyond the chain's reach, and no measured connectivity yet supplies the spectral route, so across them the coherence-to-content link is the bridge assumption E78 (Table 1). Supplemental Material gives the three derivations and the sheet's numbers.

Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i + \eta o_i$. Content and observation discrepancies bounded by ε and δ give a next discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, reduce an initial mismatch. Approximate observations leave a nondecaying term in the error bound. Taking a uniform bound on Eq. (5) connects this update to the phase process.

4.3 What approximate agreement determines

Exact gluing requires exact restrictions. For nonnegative profiles on a finite substrate, suppose every overlap discrepancy is at most ε in uniform distance.

260 Weights that vanish outside their patches and sum to one at each site select
 261 a global profile by weighted averaging. The selection lies within ε of each
 262 patch there, and any two such selections differ by at most ε , independently
 263 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 264 $L\varepsilon$ across selections.

265 At zero tolerance, the selection is the unique exact extension. At every
 266 positive tolerance there are families with no exact extension; selection then
 267 gives an approximation whose dependence on weights is bounded. Supple-
 268 mental Material gives a three-site witness attaining the bound. The mass-
 269 profile construction uses the scalar case of the content dynamics and addi-
 270 tionally requires nonnegativity of the initial profiles and observations. The
 271 general vector result supplies no identification with the spatial-measure sheaf.

272 These constructions concern declared objects: a specified cover, contents
 273 valued in a normed space, and encoders related on overlaps. They select
 274 neither the neural cover nor the decoder, and identifying their contents with
 275 perceptual content, or establishing that the identification bears on experi-
 276 ence, is a further modelling task; Section 8 sets out what a measurement of
 277 it would have to establish.

278 5 Self-representation: a state that models it- 279 self

280 5.1 The reflexive construction

281 Compatibility gives a global state. Reflexivity relates it to an encoding within
282 that state. Equip global sections with a complete metric. *Restriction reso-*
283 *nance* is the additional condition that the encoding is the state’s own restric-
284 tion to a region $\mathcal{A}$, so that the region encodes the state by being part of it.
285 Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

286 The region and readout are fixed across the state space. If F contracts and
287 fixes the section s_* formed from the local descriptions, Banach’s theorem
288 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
289 (Table 1 gives the full composition and its edges) supplies the contraction
290 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
291 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
292 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
293 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density
294 grows (Section 3), a rate about a different state from the one F acts on,
295 and no result derives a Lipschitz law for an encoding or readout from the
296 phase dynamics. Constructing a map with that factor remains a modelling

obligation. A fixed contraction can reconstruct accurately only a restricted family of states: if its error is at most ε throughout that family, any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

In the cup example, the global state includes the represented scene and the perceiver's relation to it. A local encoding must retain enough information for the readout to recover that modelled state; cup position alone would be insufficient if hand configuration and viewpoint could not be reconstructed.

For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruction gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.

315 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 316 existing self-prediction map.

317 This is an additional accuracy criterion on a family, with no requirement
 318 for multiple exact fixed points. For the same fixed contracting map used by
 319 E89, however, the diameter bound above is an additional restriction: at zero
 320 error the family contains at most one metric-distinct state. A broad family
 321 needs an input-conditioned map, one contraction F_u per input u . If F is
 322 L -Lipschitz in u , members with inputs u, v lie within $(2\varepsilon + L d(u, v))/(1 - \Lambda)$
 323 and the fixed points within $L d(u, v)/(1 - \Lambda)$, so a self-state can follow the
 324 scene. The code criterion then binds within each input: a readout storing
 325 one state per input reconstructs without encoding anything. No physical
 326 F_u is constructed, so under E89 the minimal-self correlate is a condition on
 327 one self-state. A blind encoder or constant readout fails the criterion for
 328 separated states despite having a unique fixed point.

329 Where the encoding is read as the state's own restriction to a region, the
 330 criterion becomes a condition on the region. Restriction resonance requires
 331 the encoder and that restriction to identify the same pairs of states, so a
 332 boundary reconstructing two relevant states more than 2ε apart while its
 333 region reads them identically is not resonant. Read forwards, a resonant
 334 boundary that reconstructs a pairwise separated family restricts injectively
 335 on it. Which states a region reads identically is fixed by where the region
 336 sits, before an encoding is chosen, so the resolution the family demands is
 337 the region's and no readout supplies it. The implication runs one way: a

338 region that does resolve the family is not thereby shown to carry a resonant
 339 encoding, and E89 supplies no such resolution.

340 Resonance and resolution are jointly satisfiable without a local fold. Take
 341 the region to be the whole substrate and use the identity readout: restric-
 342 tion is the identity, so this boundary is resonant and reconstructs every de-
 343 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 344 E89’s strict contraction premise on a nontrivial metric space. A properness
 345 condition on the region is therefore a separate requirement, and the fixed-
 346 contraction diameter bound remains in force when E89 is also required. With
 347 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 348 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 349 so the family is no larger than the δ -packing number of what the region car-
 350 ries. At the whole substrate it says nothing, because the region’s range is
 351 everything. The gain L is a declared input, derived here for no physical
 352 readout, and the bound weakens without limit as the gain grows.

353 **5.3 Agreement after a change takes communication**

354 Snapshot compatibility does not specify how quickly descriptions respond
 355 when a shared quantity changes. In a synchronous network whose updates
 356 read only the site’s own state and declared neighbours’ messages, two initial
 357 configurations agreeing on a site’s causal past through T rounds produce the
 358 same state there at round T . The deadline consequence is the one Section 2
 359 states, and it follows from the reconstruction bound in Section 5.2: a report

360 from that site is a readout of a code that the two interventions leave identical.

361 A three-site delay line witnesses failure after one round and exact report-
362 ing after two; cutting the path prevents delivery at every round. A report
363 may still happen to be correct about one fixed world. The bound concerns
364 responsiveness across interventions.

365 Throughout this section *prediction* names a reconstruction map, and the
366 relevant family of states, the metric, the tolerance, the communication graph
367 and the deadline are all declared inputs, as are the represented variables and
368 the fidelity counted as accurate. Reading the map as temporal neural pre-
369 diction would require dynamics; reading the resulting code count as physical
370 memory capacity would require a hardware model, which a state-dependent
371 readout would have to extend.

372 6 What the conditions ask of a digital candi- 373 date

374 A claim that a digital system, such as a large language model (LLM) run-
375 ning on graphics processing units (GPUs), has unified content or represents
376 itself commits its maker to the measurements this account asks of cortex.
377 The maker must identify the shared quantities the system’s parts describe,
378 show that decoders fitted independently on those parts agree about them,
379 and show that a declared local code separates the relevant family of states
380 and survives intervention. The indicator-properties approach assesses arti-

381 ficial systems against computational properties drawn from theories of con-
 382 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
 383 a measurement rather than an architectural property. A digital system is
 384 also where they can be measured today, with full access to its state and
 385 exact interventions on it: fitting decoders for one shared quantity on two
 386 separate layers, heads or token positions, and testing their agreement and
 387 whether reconstruction survives intervention, would carry out the proposed
 388 measurement directly.

389 One constraint follows from the architecture alone. Within one forward
 390 pass of a causally masked model, position t reads only its causal past, so an
 391 intervention outside that past cannot be reconstructed at t by that step. This
 392 is the deadline result of Section 5 applied to the mask, and it bounds which
 393 variables a position can encode about the rest of the sequence within one
 394 step. A second constraint needs a premise the architecture does not supply:
 395 a linear encoder and readout factoring through d coordinates pay reconstruc-
 396 tion error at least the discarded spectral tail of the coupling operator, but
 397 query/key dimension does not supply such a bottleneck for softmax atten-
 398 tion. These results rank no architecture family, imply no general inferiority of
 399 GPU hardware and settle no question of machine experience. Supplemental
 400 Material gives the bottleneck derivation, a synthetic two-view task with its
 401 controls, the information ceiling of token outputs and the hardware scope.

402 **7 The formal composition**

403 The Lean development composes the three requirements into one conditional
404 theorem. Its passive and feedback branches each accept eight connecting hy-
405 potheses, running from finite capacity through a register, dissipation and a
406 continuum coupling to the field, coherence, the cover and the self-prediction
407 fixed point. Each edge is supplied rather than derived from the one above it,
408 and the inferential content of the composition is its last step, where contrac-
409 tion and a fixed glued section give uniqueness. Table 1 lists the four edges the
410 article argues from; Supplemental Material gives the full chain, its thermo-
411 dynamic edges and the register ledger of the feedback branch. Both branches
412 also take a cover carrying exact overlap compatibility and a reflexive bound-
413 ary specifying encoding and readout, and conclude that the glued section is
414 the unique fixed point of self-prediction. Composing the edges establishes
415 that the premises are consistent, named in one place and machine-checked
416 together; it does not establish that self-representation follows from coher-
417 ence. Joint witnesses satisfy every edge of each branch in a specified model,
418 and establish no common cortical mechanism. The development declares no
419 additional axioms beyond its libraries' foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover’s nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve’s diameter; where the configuration is frequency-locked it carries through the coupling graph’s algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

420 8 What the observations establish

421 The composition holds under its premises. Whether observation can estab-
422 lish those premises is a separate question, and the three requirements it
423 connects concern different objects: phase dynamics, local descriptions and
424 internal encodings. Their observables must retain these distinctions. The
425 following controls show which mechanisms remain indistinguishable, which
426 temporal approximations fail, which states one observable cannot separate,
427 and what independent content measurements must establish. Each is a finite
428 numerical regime on a declared model, with full parameters and controls in
429 Supplemental Material, and none of them calibrates a cortical quantity.

430 8.1 Stationary distributions leave the mechanism un- 431 determined

432 Equation (2) gives a concentration–coherence relation with no fitted shape
433 parameter. Estimate r from circular resultant length and a from density
434 shape: inverting R from the measured r would make agreement automatic.
435 Even independently estimated agreement identifies a statistical family shared
436 by different mechanisms.

437 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
438 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
439 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
440 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

441 and diffusion D_0 have that same density. Their stationary distributions co-
 442 incide window by window despite different causes.

443 Resolved dynamics supply additional information. In the ideal model
 444 with continuous latent paths, quadratic variation accumulates at rate $2D$
 445 and conditional increment means constrain drift; a filtered observation of
 446 the phase needs its own observation model before either is read off. Separat-
 447 ing endogenous coupling from a matched common drive additionally requires
 448 a measured drive or controlled perturbation; identifying electromagnetic cou-
 449 pling requires evidence specific to that interaction. Supplemental Material
 450 gives the equivalence construction and its numerical check. Phase statistics
 451 pooled over time and over bipolar pairs cannot supply these estimates: on
 452 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-
 453 fectly phase-locked signal at $r = 0.0001$, so a spatial test needs an estimand
 454 that removes the common rotation and a montage whose effect on spatial
 455 phase is modelled; Supplemental Material reports this estimator exercise and
 456 the dependence-aware design.

457 **8.2 Finite ramps change the observable onset**

458 The square-root law in Eq. (3) describes stationary solutions, and a chang-
 459 ing coupling needs time for the phase distribution to approach them. In a
 460 deterministic mean-field control crossing the threshold, none of the scanned
 461 rates meets the declared tracking tolerance at onset, so a slow control vari-
 462 able supplies no general guarantee of branch tracking. In a finite-population

463 ramp study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps de-
 464 lay escape. The fitted delay exponent 0.444 has a replica-bootstrap interval
 465 $[0.398, 0.496]$ that excludes the ideal $1/2$, a deterministic reference shows that
 466 the escape criterion changes the fitted exponent, and no common asymp-
 467 totic exponent is established. Delayed passage through a noisy bifurcation
 468 (Berglund & Gentz, 2002) makes the apparent exponent depend on ramp
 469 speed and fitting window. Supplemental Material gives the sweep and its
 470 figure, the fluctuating-coupling and population-size controls, and the signed-
 471 network crossing bracket.

472 **8.3 A winding state leaves the order parameter am-** 473 **biguous**

474 The global order parameter r discards spatial arrangement. On a periodic
 475 sheet, a phase field can wind once around the sheet and have $r = 0$ while
 476 neighbouring sites remain locked. At identical frequencies, every such wind-
 477 ing is stationary for an isotropic symmetric kernel: contributions at opposite
 478 separations cancel. It violates the quarter-turn initial-spread premise of the
 479 convergence result, which is why that result can still select the uniform state.

480 In the spatial control, the winding persists below the interpolated de-
 481 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 482 and winding starts under the same coupling, diffusion and kernel end at
 483 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-

484 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 485 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 486 realizations. These are finite controls, not a cortical wave identification or a
 487 derived transition.

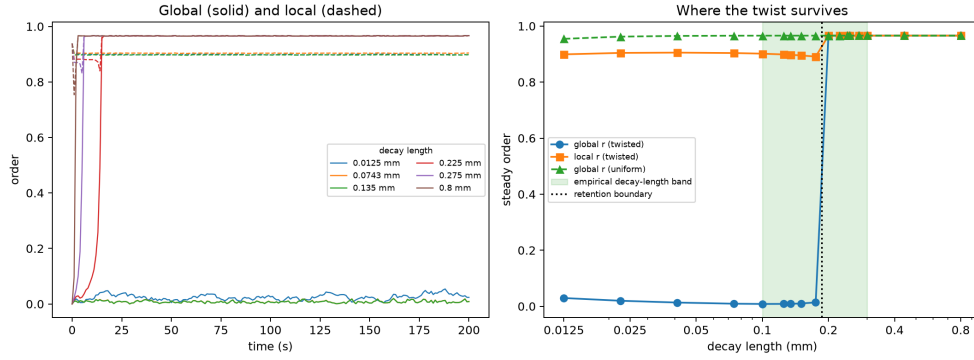


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

488 Linearization about a winding gives explicit mode rates, which concern
 489 linear stability rather than the measured retention boundary. An amplitude
 490 extension adds an existence condition under which a broad kernel can remove
 491 a winding that the phase-only equations retain as a stationary state.

492 Neither model turns these states into measured cortical spirals. The pe-
 493 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has

no propagation speed, and the amplitude variable has no calibrated relation to a recorded envelope. Supplemental Material gives the spectral, amplitude and defect results. Locally, Eq. (5) can still bound content disagreement on a winding patch when shared encoders are supplied, although its global $r = 0$ form is uninformative. A nonzero ring winding number q also obstructs a global real-valued phase lift; it has no identified content meaning here. Thus an onset criterion based only on r can register a change of basin as well as a change of coupling. The protocol of Section 9.3 needs a spatial phase observable.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling within clusters. Supplemental Material proves descent of this objective for a specified gradient flow at fixed phases and distinguishes it from entropy production.

The finite control co-evolves noisy phases and symmetric, nonnegative

couplings at fixed total resource. Its template is a diagnostic covariance of latent signals, never a training input. Across three illustrative seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and in-

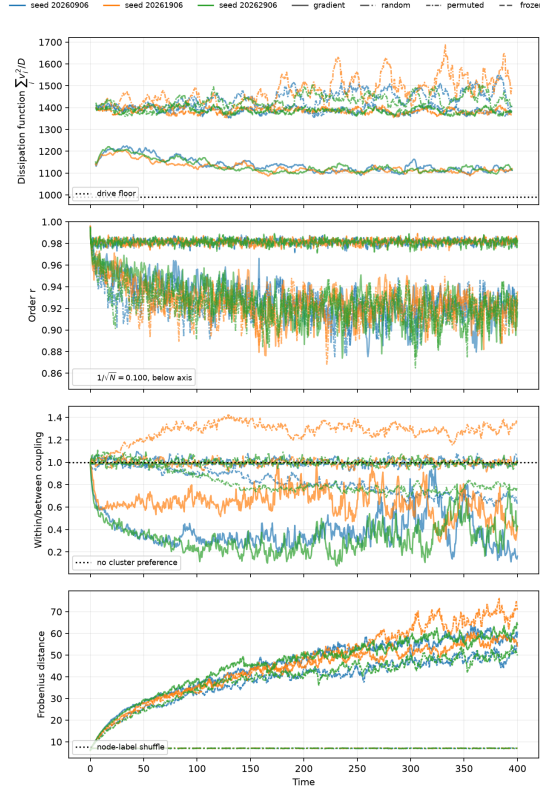


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

compatible examples. An implementation on synthetic data separates these cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. Section 4 gives the decodability rule that fixes the cover without consulting agreement.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what

558 observations must distinguish before a phase trace can support this identifi-
559 cation.

560 **9.1 Physical identification and calibration**

561 The cortical hypothesis identifies part of effective coupling with endogenous
562 extracellular electromagnetic interactions. Fields can modulate spike timing
563 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
564 propose coordination of distributed neural information (Pinotsis & Miller,
565 2022; McFadden, 2020). Phase organization under propofol motivates study-
566 ing this regime (Bastos et al., 2021), but does not measure coupling.

567 The formal field predicate requires a jointly continuous kernel, a normal-
568 ized substrate, coupling equal to the kernel average, and no single site pro-
569 viding more than half that strength. Toy constructions satisfy it. A cortical
570 realization must specify the scale at which the kernel and its reduction ap-
571 ply: calcium imaging reports different state-dependent modularity patterns
572 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

573 Cortical columns are a serious candidate for the oscillator index. They are
574 delineated by response properties and anatomy rather than by the agreement
575 they would be used to establish (Mountcastle, 1997), and a phase can be
576 estimated from a column-sized territory at the electrode spacings used for
577 cortical wave measurements (Townsend et al., 2015). They are a poor content
578 cover: non-overlapping columns form a disjoint cover (Section 4), much of
579 association cortex has no accepted columnar parcellation, and the column’s

standing as a canonical unit is contested (Horton & Adams, 2005; Rakic,
 2008). The content cover is the decodability cover of Section 4, and the
 bridge assumption E78 records its identification with the oscillator index.

The spatial reduction is also testable. A row-sum aggregation at zero
 diagonal tracks the uniform-coupling control over the tested spatial decay
 range, while frequency heterogeneity displaces the operational threshold. The
 fit window biases both estimates relative to $2D$; agreement with the control
 is therefore the supported conclusion. Supplemental Material reports the
 reduction, its numerical error and the distinct assumptions needed to identify
 the kernel average with the scalar coupling of the installed-energy condition.

The field commitment requires physical evidence beyond a choice of math-
 ematical representation. The support comparisons in Supplemental Material
 concern specified support constraints and supply neither that evidence nor
 energy prices.

A spike-time shift alone supplies no coupling rate. The dimensionless
 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

This rate describes how induced timing shifts accumulate into drift. Nei-
 ther it nor the installation factor κ follows from a heat bound. Testing the
 physical threshold requires independent estimates of the interaction kernel,

its aggregation and diffusion in the same cortical regime.

9.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (10) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives a deterministic tracking approximation that fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

This is a deterministic tracking approximation from Eq. (3) at fixed D ; the crossing controls in Section 8.2 show that it fails as an observed-onset ap-

proximation under the tested initialization. A separation of geometry and phase time scales can support tracking away from threshold. The awakening hypothesis needs a calibrated dynamical prediction for an operational onset above the finite-size fluctuation floor. Crossing K_c alone does not supply a recovery latency.

9.3 A protocol for the awakening experiment

Neural inertia describes a behavioural barrier and hysteresis between anaesthetic induction and emergence (Friedman et al., 2010); recovery can pass through metastable activity states (Hudson et al., 2014; Proekt & Hudson, 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field mechanism from a synaptic one. A possible discriminator is spatial onset relative to independently estimated local field coupling and network connectivity. The present mean-field model has no spatially varying κ and derives no nucleation order, so a field-based onset map needs a separate model specified before the observations are inspected.

For that comparison, partition the recording into prespecified regions i and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements, and a connectivity score c_i from a separately estimated structural network. For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

641 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

642 Predicted onset is the first crossing of a region-specific threshold fixed from
643 pre-onset fluctuations. Fit the two scales on training sessions, then com-
644 pare held-out onset times and site order under the same observation model.
645 Larger field scores predict earlier onset only when baseline order and diffu-
646 sion do not reverse the ranking. This declared approximation omits interre-
647 gional coupling and saturation; a successful ranking would support its field
648 covariate, not prove ephaptic causation. Supplemental Material specifies the
649 controls and failure conditions.

650 A within-subject protocol can manipulate emergence rate, with spatially
651 resolved recordings to compare the proposed field and connectivity models:

- 652 1. **Manipulate emergence rate.** Vary the propofol infusion taper
653 across sessions within the same subject, recording drug concentrations
654 or a validated pharmacokinetic proxy to quantify the manipulation.
- 655 2. **Measure behavioural recovery.** Record time-to-response to a stan-
656 dard stimulus as the independent behavioural endpoint, rather than
657 estimating ΔK directly.
- 658 3. **Resolve spatial onset.** Identify anatomical locations where coher-
659 ence first crosses the baseline-scaled threshold and score the two maps

660 of Eq. (12) on held-out sessions. The two covariate maps are fixed
661 before the outcome sessions are inspected.

662 **4. Match physical controls.** Confirm that the observed coherence-to-
663 amplitude relation holds under the specific local diffusion D and ob-
664 servation noise. The model’s quadratic-variation rate $2D$ applies to
665 the latent phase; a band-limited analytic phase is smooth below its
666 correlation time, and on a constructed field its squared-increment rate
667 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
668 fore requires an explicit observation model for filtering and sampling.

669 Predeclare the recording window, regional fluctuation thresholds, and a
670 shared timing-error model. Fit the two scales and that observation model
671 on training sessions only; on held-out sessions, score the observed site-onset
672 times and sites that never cross by the same regional predictive rule. Report
673 site order as a secondary outcome and behavioural time-to-response sepa-
674 rately. Sessions without a crossing are censored observations, not zero delays
675 or exclusions. Compare the two scores only where independently estimated
676 field and connectivity maps predict distinguishable onset patterns at the
677 recording resolution. A held-out field-model advantage would support that
678 specified spatial predictor; it would not identify an ephaptic cause, shared
679 content or experience. Supplemental Material gives the controls and pilot
680 quantities needed to judge whether the comparison is informative.

681 10 Discussion

682 10.1 Unity and a minimal self as interpretive motiva- 683 tion

684 We propose the state glued from compatible local descriptions as a formal
685 correlate of conscious unity, and its accurate internal reconstruction as a cor-
686 relate of a minimal self. In the cup example, this would mean one description
687 that preserves the cup’s relation to the perceiver as well as the scene. Com-
688 patibility permits distinct contents and uncertainty; the shared quantities
689 must be chosen from independent evidence about what each region encodes.
690 The reconstruction need not be a verbal or autobiographical account.

691 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
692 ing the subject of an experience, present before any narrative or reflective
693 self-description (Gallagher, 2000). Two of its features motivate the proposed
694 correlate. It is not a separate object of experience but a perspective built
695 into the experienced scene; correspondingly, the reconstruction is not a sec-
696 ond representation beside the glued state but a map whose fixed point is
697 that state, read through a part of it. It is immediate rather than inferred
698 from a higher-order state; correspondingly, the fixed point needs no further
699 state that represents the first. The correspondence is structural and does not
700 show that a fixed point is experienced as a perspective. Self-model theory
701 locates the phenomenal self in a transparent model a system has of itself
702 (Metzinger, 2003), and interoceptive-inference accounts ground it in predic-

tion of the body’s own states (Seth, 2013). The present condition is narrower
 than either: it asks only that a declared local encoding recover the global
 state, including perceiver-relative variables, at a declared fidelity, and it says
 nothing about transparency or about which bodily variables must be mod-
 elled. Either account could supply the family of states and the fidelity that
 the condition leaves as inputs. Under E89 that family is one state at zero er-
 ror and a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Section 5.2),
 so under E89 the condition constrains a single self-state; a self that per-
 sists through change needs an input-conditioned map, for which no physical
 readout is given. Its proposed relation to experienced perspective remains
 an interpretation. Neither identification is among the connecting hypothe-
 ses of the composition (Table 1), and satisfying them would not measure
 experience. The unity proposal is also only as definite as its cover. The
 gluing theorem holds for whatever cover it is given, a disjoint cover satisfies
 it trivially, and the cover of interest overlaps in content rather than in space.
 Section 4 proposes fixing that cover by decodability on held-out data, which
 excludes covers chosen for agreement and makes non-degeneracy a testable
 property of the recordings. Its variables, criterion and decoder class are pre-
 registered by rules that never consult agreement, and the formal results are
 stated for spatial covers.

723 10.2 A comparison that could disconfirm the link to 724 content

725 The framework predicts that conscious content tracks compatibility, not co-
726 herence. The comparison must also separate compatibility from how well
727 each region encodes the stimulus, because two accurate decoders agree auto-
728 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
729 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

730 Consciously perceived stimuli are usually decoded better, so higher agreement
731 on those trials would follow from single-region fidelity alone. At matched
732 single-region error, agreement differs only through the covariance of the two
733 errors, which is the term compatibility bears on: two regions describing one
734 content should err together.

735 Present matched stimuli that are perceived on some trials and not on
736 others, for example at masking or rivalry threshold, and fit decoders for one
737 shared stimulus quantity independently on two regions that both encode it,
738 selected by the decodability criterion of Section 4 on separate data. Clas-
739 sify trials by perceptual state with a no-report measure, such as reflexive
740 eye movements or pupil responses that track the dominant percept in rivalry
741 (Tsuchiya et al., 2015), so that the classification does not also select for the
742 neural correlates of reporting; report-based classification is a secondary anal-

743 ysis. Match the two classes on phase coherence between the regions and
744 on each region’s own decoding error (or, where the stimulus value is uncer-
745 tain, on each region’s signal strength). The statistic is the excess agreement
746 over a fidelity-matched null: pair region *A*’s decode on one trial with region
747 *B*’s decode on another trial of the same stimulus and error stratum, which
748 preserves both marginal accuracies and removes any coupling of the errors.

749 The proposed link predicts a larger excess on perceived trials at matched
750 coherence and matched single-region accuracy (Fig. 5). Equal excess across
751 classes, or excess that follows coherence and not perceptual state, counts
752 against it. A positive result is not yet specific: noise shared upstream of both
753 regions also correlates their errors, so the excess should be compared against
754 an upstream region’s decoding error, matched in the same way. A self-model
755 comparison has the same form: at matched accuracy for the scene variable,
756 the reconstruction of perceiver-relative variables, such as hand position rel-
757 ative to the target, from a declared local encoding should be more accurate
758 on perceived trials. These are correlational tests of proposed correlates, and
759 the no-report classification is itself an indirect marker of experience.

760 10.3 What a test of the full proposal must establish

761 A test needs separate estimates of effective coupling and diffusion, shared
762 content on overlapping regions, and reconstruction from a specified encoding
763 at a declared fidelity. The cover, relevant state family and decoder must
764 be fixed without selecting them for agreement. An independent experiential

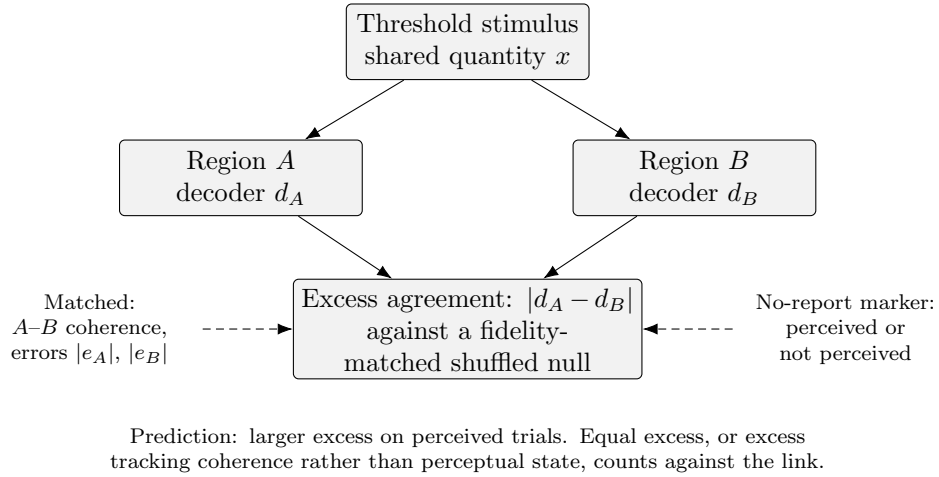


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

765 endpoint is needed to evaluate the interpretation; absence of a report alone
766 does not define absence of experience. A unified episode with incompati-
767 ble decoded descriptions would challenge the unity identification. A unified
768 episode lacking the proposed self-encoding would challenge the minimal-self
769 claim. Conversely, satisfying the mathematical conditions without such an
770 episode would challenge their sufficiency as correlates.

771 For the physical route, cortical actuator modes and prices, the geometry–
772 coupling relation and a dynamical onset model remain to be measured. The
773 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
774 diction. Controlled changes in coordination and independently decoded con-

775 tent agreement would address the representational connection. The synthetic
776 decoder exercise calibrates part of these measurements, and the EEG exercise
777 identifies what a pooled bipolar estimand cannot measure; neither identifies
778 a neural content code or tests experience.

779 10.4 Explanatory scope

780 Integrated-information, global-workspace and predictive-processing accounts
781 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
782 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
783 relations among coupling, local content and reconstruction. Its formal field
784 condition requires a continuous kernel, a normalized substrate, a specified
785 average coupling and noise, and a site-contribution bound. Those conditions
786 do not identify an electromagnetic substrate; the extracellular interpretation
787 needs the separate calibration of Section 9. The joint witness shows that
788 the formal hypotheses can hold together in a declared model. A cortical
789 realization and an experiential interpretation require additional evidence.

790 Four limitations bound every claim above. Synchrony is not itself an
791 experiential measure, and nothing here establishes a consciousness criterion:
792 the conditions are proposed correlates to be tested against report or another
793 independent endpoint. The present EEG analysis checks a measurement
794 pipeline and does not validate any experiential claim. The formal results are
795 conditional on declared encoders, covers and maps whose cortical counter-
796 parts are unmeasured. The cover on which compatibility is assessed is fixed

797 only relative to pre-registered variables, decoder class and criterion, and the
798 formal results are stated for spatial covers rather than for the content cover
799 that criterion yields.

800 The remaining explanatory question is whether these relations occur in
801 neural activity and matter for experience. The framework identifies candidate
802 correlates and tests of their physical and representational premises. It does
803 not explain why satisfying them would be accompanied by experience.

804 11 Conclusions

805 Phase coherence, compatibility and self-reconstruction are separable condi-
806 tions. Coherence constrains shared content only through encoders shared on
807 overlaps and with a bound that grows as the square root of the population
808 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch
809 order; chaining it through overlapping patches reaches 0.312 mm of that
810 sheet, and a spectral route removes the population count only for a declared
811 connectivity at a frequency-locked configuration. Across the centimetre sep-
812 arations of distinct cortical territories, the positive link from coherence to
813 content is the bridge assumption E78 rather than a derived result. Com-
814 patible local descriptions, not synchrony, determine one global state; and a
815 contracting self-map fixes that state while reconstructing only a family of
816 bounded diameter. Each result replaces an assumption about synchrony by
817 a measurement: independently decoded agreement on overlaps, a declared

818 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
819 ison of Section 10.2 states how such measurements could count against the
820 proposed link to conscious content.

821 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits

Table 2 (continued)

Symbol	Defined	Meaning
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	Section 9	Extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

822 Data and code availability

823 The Lean development, simulation scripts, tests and saved numerical sum-
824 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
825 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-

lib commit 5ce203ecd53f8c3e47ee5530d7c86b94a55f0183. Publication
simulation values are generated from saved summaries without rerunning
production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
and selection are described in Supplemental Material and implemented in
simulations/empirical_collapse.py. The repository contains the repro-
ducible analysis and does not redistribute the raw recordings.

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Use of AI tools

AI assistance to this work was substantial and is not separable by task.
Several coding assistants and chat interfaces were used in an interleaved way
throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them

845 they contributed to the Lean development, the simulation and analysis code,
846 and the drafting and revision of both documents. Because the tools were
847 used in combination and in alternation, no section, proof, figure or result is
848 attributable to any one of them, and none is claimed to be. The author is
849 responsible for all content, including everything a tool produced. Machine
850 checking establishes the stated Lean results under their hypotheses; it does
851 not validate biological interpretations or replace scientific review of generated
852 code and prose.

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